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Conducting reproducible ecosystem modeling using the open source mass balance model Rpath

Sean M. Lucey*,[a], Sarah K. Gaichas[a], Kerim Y. Aydin[b]

[a] NOAA, National Marine Fisheries Service, Northeast Fisheries Science Center, 166 Water Street, Woods Hole, MA 02543, United States
[b] NOAA, National Marine Fisheries Service, Alaska Fisheries Science Center, 7600 Sand Point Way NE, Seattle, WA 98115, United States

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ABSTRACT

Ecosystem models are important tools for conducting ecosystem-based management. A particularly useful method of characterizing the flow of energy through an ecosystem and the subsequent direct and indirect implications of management actions is mass balance modeling. Here we outline the equations as utilized in Rpath, an R implementation of the mass balance algorithms popularized by Ecopath with Ecosim that are designed to work with fisheries data sources. We believe that common practices in R will aid in the reproducibility of conducting analysis using a mass balance model as all of the code is contained within a single script file. This includes the built-in statistical and graphical functions of R. In addition to added reproducibility, R is a coding language with which ecologists are familiar. This familiarity offers greater flexibility for practitioners to tailor the model to their needs. We have made the code available on an open software development platform which should aid in continuous community development of the tool.

1. Introduction

Mass balance models describe the structure and flow of energy through the system. They are commonly used for quantifying food web interactions in a "whole-ecosystem" context. The most popular of these models is Ecopath which was first developed by Polovina (1984) to explain the net primary production required to support a tropical reef system. That model has been modified and expanded to fit all types of aquatic and even terrestrial ecosystems (Fretzer, 2014). Part of its power is that it makes use of input quantities that are regularly measured in single-species resource management, such as biomass, mortality rates, and growth rates, that are then placed into the mass-balance framework to determine ecosystem "unknowns" such as lower trophic-level productivity.

Ecosim (Walters et al., 1997) uses Ecopath models as a basis for parameterizing a set of biomass-dynamics differential equations that include functional responses between predators and prey, removals by humans (hereafter "fishing"), and environmental drivers that may be time-varying. Ecosim is used to conduct dynamic simulations, either to explain historical changes in ecosystem structure and function, or to predict the direct and indirect effects of both human-driven and natural ecosystem perturbations. A later iteration of the Ecosim equations (Walters et al., 2000) added a method for modeling age-specific life stages (hereafter "stanzas") for individual species using delay-difference equations in conjunction with the biomass dynamics model, to explicitly model recruitment, variable growth, ontogenetic diet shifts, and age-based fisheries selectivity. Ecopath and Ecosim are used together in an ecosystem-based management context, to identify and quantify major energy flows in the system, describe ecosystem resources and their interactions, evaluate the ecosystem effects of fishing or environmental changes, and exploring management policy options (Plaganyi, 2007).

A large measure of these tools' popularity comes from the development of a user-friendly package, Ecopath with Ecosim (EwE; Christensen and Pauly, 1992; Steenbeek et al., 2015). EwE has many strengths including ease of use through a convenient graphical user interface (GUI) and the ability to track data quality as an inherent part of EwE data entry. Further, the EwE package includes a wide range of add-on tools for calculating ecosystem properties such as cycling, diversity, and throughput. Users can input their parameters directly into a spreadsheet or import them from a comma delimitated file. Modifications can be made within the GUI for running simulations. However, this strength is also a weakness. Users must take extreme caution to insure that user error is not introduced as running multiple simulations requires users to record the settings for each run. It is possible that some modifications may not be saved and therefore limit the reproducibility by other users.

EwE software is open source and freely available at www.ecopath.

* Corresponding author.
E-mail address: Sean.Lucey@NOAA.gov (S.M. Lucey).

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org. Open source means that there is universal access to the code and that the code can be redistributed or subsequently improved by any interested users (Steenbeek et al., 2015). This allows users to overcome some GUI-specific issues such as running scripts directly from the base code. The base code is built on the Microsoft.NET platform, primarily using Visual Basic.

Many ecologists are familiar with the rapidly growing statistical language R (Tippmann, 2014). It is the software of choice for many scientists who are transitioning from costly, platform-specific commercial software packages. A recent search on Elsevier's Scopus database showed that one in 100 scholarly articles cite R or one of its packages, with even greater citation rates in the agricultural and environmental sciences (Tippmann, 2014). R is available for multiple operating systems and has a rich user community with strong on-line support. R was the 12th most popular programming language on GitHub, a popular version control repository, including the most new collaborations per repository which demonstrates that it is being actively used and modified (Smith, 2015). R also boasts a publication quality graphical engine that allows users to generate figures in the same programming environment that they conducted their statistical analyses.

Here we introduce an R implementation of Ecopath and Ecosim called Rpath, which is meant to be a complement to the existing EwE software. Our goals were to develop a flexible tool that would take advantage of the built-in graphical and statistical prowess of R and allow for better reproducibility of ecosystem modeling by capturing all changes in a script format meaning analyses can be easily recreated by others. Furthermore, recoding of Ecopath and Ecosim using R's extensive libraries for linear algebra and matrix manipulation allows for greater readability of the underlying code and greater extensibility by the community of researchers familiar with R.

Rpath includes the mass-balance methods of Polovina (1984), the biomass dynamics methods (Walters et al., 1997) and the life history stanzas methods (Walters et al., 2000). The mass-balance algorithms are implemented using R's linear algebra library. The core dynamic simulation derivative and numerical integration routines are implemented using C++ code based on Aydin et al. (2007) re-written using the Rcpp library (Eddelbuettel and Francois, 2011) to make use of R's vector operations and transparent coding structure within the C++ code, while allowing the parameter setup and results analysis to be performed seamlessly in the R environment (the Rsim family of functions within the Rpath package).

This paper is not meant to serve as a user's manual for Rpath but rather to describe the mass balance and dynamic simulation algorithms that are implemented in the base library of Rpath. The R package, which can be downloaded at https://github.com/NOAA-EDAB/Rpath, includes several vignettes that demonstrate how to run the model.

2. Mass balance in Rpath

2.1. Processes of the function Rpath

Ecopath is a system of linear equations that describes the mass balance of an ecosystem. There are two master equations that govern the mass balance, one that characterizes production and the other consumption. Production is calculated as the sum of all predation on a group as well as non-predatory losses (fishing, emigration, etc.) while consumption must account for all production as well as unassimilated food and respiration. The production equation can be rearranged as Eq. (1) (see Christensen and Pauly, 1992 for full derivation). The parameters for ecopath equations are described in Table 1 with subscript definitions in Table 2.

$$B_i PB_i EE_i = \sum_{j=1}^{n} B_j QB_j DC_{ij} + EX_i \tag{1}$$

This form of the Ecopath master equation allows for the algebraic solution for unknown biomasses (B) or ecotrophic efficiencies (EE). Rpath unlike the EwE software does not solve for unknown PB or QB. On the right hand side of Eq. (1) is the sum of predator mortality as well as other exports (EX) from the system. These exports include fishery removals, biomass accumulation, and net migration. Within the Rpath package, mass balance is carried out with the Rpath family of functions (Fig. 1) with the function *rpath* executing the solution of the system of equations. First, it simplifies the list of possible unknown parameters using relationships to other known parameters. Specifically, unknown growth efficiencies (GE, alternately named PQ in some literature for production to consumption) are solved using Eq. (2).

$$GE_i = \frac{PB_i}{QB_i} \tag{2}$$

After the list of unknown parameters is minimized, the remaining unknowns are solved using matrix multiplication. Eq. (1) is expressed in the traditional matrix algebra form of $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is an n x n matrix with the diagonal of coefficients either $B_i PB_i$ or $PB_i EE_i$, based on which is the unknown parameter, and the off-diagonal coefficients relating to consumption if the biomass of a predator is unknown, $\mathbf{x}$ is an n-dimensional vector of unknown B or EE, and $\mathbf{b}$ is an n-dimensional vector of mortality terms. Eq. (1) is solved by inverting $\mathbf{A}$ using the general inverse function (*ginv*) from the MASS package (Venables and Ripley, 2003) and multiplying by $\mathbf{b}$.

In addition to calculating unknown B or EE, Rpath calculates several ancillary parameters. These include detrital biomass and ecotrophic efficiency, trophic levels, and mortality terms. Detrital biomass and ecotrophic efficiency are not necessary for Ecopath but are necessary when transitioning to the dynamic simulation Ecosim. Trophic levels and mortality terms are of general interest to researchers.

Typically, detrital biomasses are unknown. While detritus groups do not respire or consume anything, detritus is consumed in most models. Therefore, it is important to account for how much detritus exists and is used. Detritus is generated through non-mortality loss, the unassimilated portions of consumption, and fishery discards Eq. (3). It is subsequently removed via consumption Eq. (4). Knowing both detrital input and output, EE_{dt} is calculated as $detoutput_{dt}/detinput_{dt}$. Further if detrital biomass is not provided, it is calculated using $B_{dt} = detinput_{dt}/PB_{dt}$. Note if PB_{dt} is not provided, a default value of 0.5 is used, so the default detrital biomass is twice $detinput_{dt}$.

$$detinput_{dt} = \sum_{j=1}^{n} \left(detfate_{dtj} \times (M0_j B_j + B_j QB_j U_j) \right) + \sum_{k=1}^{n} \left(detfate_{dtk} \times \sum_{j=1}^{n} C_{jgdisc} \right) \tag{3}$$

$$detoutput_{dt} = \sum_{j=1}^{n} B_j QB_j DC_{dtj} \tag{4}$$

The trophic levels for ecosystem components are calculated in a similar method to unknown B or EE using matrix algebra. The trophic level equation Eq. (5) is rearranged as Eq. (6). In this case, the matrix notation $\mathbf{b}$ is an n-dimensional vector of ones. The $\mathbf{A}$ matrix is a transposed matrix of negative diet composition in the off-diagonals, with ones in the diagonal. Rpath then solves for $\mathbf{x}$, an n-dimensional vector of unknown trophic levels. In the case of mixotrophs, their diets are renormalized to take into account the proportion of primary production as, $DC'_{ij} = DC_{ij} \times (1 - \%PP_i)$.

$$TL_j = 1 + \sum_{i=1}^{n} TL_i DC_{ij} \tag{5}$$

$$\left(1 - \frac{\sum_{i=1}^{n} TL_i DC_{ij}}{TL_j}\right) TL_j = 1 \tag{6}$$

Table 1
Pameters used in Rpath equations.

Symbol	Units	Parameter	Description
B	$t\ km^{-1}$	Biomass	Amount of biomass within the modeling domain
PB	yr^{-1}	Production/Biomass	Rate of biomass growth per unit of biomass
QB	yr^{-1}	Consumption/Biomass	Consumption rate per unit of biomass
EE	proportion	Ecotrophic Efficiency	Fraction of a prey species' annual production accounted for within the model
GE (PQ)	dimensionless	Gross Food Conversion Efficiency	The ratio of production to consumption
DC	proportion	Diet Composition	Percentage of predator's diet comprised of a particular prey
EX	$t\ km^{-2}\ yr^{-1}$	Exports	Simplified notation for the sum of biomass accumulation and net migration. Derived from the master Ecopath equation.
M0	yr^{-1}	Unexplained Mortality	Rate of mortality that is not explained within the model
U	proportion	Unassimilated Food	Percentage of consumed food not utilized for growth
Res	proportion	Respiration	Percentage of consumed food lost to respiration
detfate	proportion	Detritus Fate	Percentage of flow to particular detrital groups
detinput	$t\ km^{-2}\ yr^{-1}$	Detritus Input	Amount of detritus created within the system
detoutput	$t\ km^{-2}\ yr^{-1}$	Detritus Output	Amount of detritus utilized within the system
TL	dimensionless	Trophic Level	Position occupied in the food web starting at 1 for producers
M2	yr^{-1}	Predation Mortality	Mortality rate due to consumption by predators
F	yr^{-1}	Fishing Mortality	Mortality rate due to fishery removals
C	$t\ km^{-1}$	Catch	Amount of biomass caught by a fleet
q	dimensionless	Catchability	The effectiveness of a fleet to catch a group based on an initial effort level of 1
E	yr^{-1}	Fishing Effort	Amount of effort exerted by a fleet scaled to an initial effort level of 1
PP	proportion	Primary Production	How much of a species energy is derived via photosynthesis

Table 2
Description of subscripts used.

Subscript	Description
i	Species or group of interest
j	Interacting species with i
dt	Detrital group
g	Fishing fleet
land	Landings
disc	Discards
t	Timestep
a	Age
s	Stanza
ms	Multi-stanza group
leading	Leading stanza group
spawn	Spawning
∞, max	Maximum
0, min	Minimum
m	Month
net	Net Total

The final set of ancillary calculations is the various mortality rates on a species group. These include predation mortality(*M2*), fishing mortality (*F*), and "other" mortality (*M0*). Predation mortality is calculated as the loss of biomass to a predator divided by the biomass of that prey group Eq. (7). Total catch for each species *i* and fleet *g* includes both landings ($C_{ig,land}$) and discards ($C_{ig,disc}$); landings and discards are entered separately for each fleet *g; F* for each species is the sum of all catches of that species divided by its biomass Eq. (8). Discards are considered an input to a detritus group as 100% mortality is assumed. "Other" mortality accounts for losses outside of the model domain calculated as Eq. (9).

$$M2_{ij} = \frac{DC_{ij} QB_j B_j}{B_i} \tag{7}$$

$$F_i = \frac{\sum_{g=1}^{n} C_{ig,land} + C_{ig,disc}}{B_i} \tag{8}$$

$$M0_i = PB_i(1 - EE_i) \tag{9}$$

2.2. Multistanza groups

To account for ontogenetic shifts in species diet and mortality rates, Rpath includes the ability to parameterize multistanza species groups. Linking these groups together by simple biomass flow would produce unrealistic dynamics (Walters et al., 1997). As a result, a delay difference model is parameterized so that flow between stanzas is based on numbers and biomass. Although this is more of a concern for dynamic simulations, Rpath bases the biomass and consumption estimates for all of the stages based on one leading stage. The function *rpath.stanzas* will calculate the appropriate fields by calculating the weight at age using the generalized Von Bertalanffy growth equation (Eq. (10); Essington et al. (2001)). The more common specialized growth equation assumes $d = 2/3$ which creates a carrying capacity parameter, *Ksp*, that is $k \times 3$. EwE uses the specialized version of the growth equation so for consistency, *rpath.stanzas* takes *Ksp* as the parameter and converts to *k* within the code. As with the general Ecopath equations, parameter definitions for stanza equations can be found in Table 3.

$$W_a = W_\infty (1 - e^{-k(1-d)(t-t_0)})^{\frac{1}{1-d}} \tag{10}$$

Assuming a stable age distribution, *B* and *QB* for all non-leading stanzas are calculated using the proportion of the population at age *a* as Eq. (11), where l_a is the population growth-rate-corrected survivorship (Eq. (12); Christensen and Walters (2004)). The relative biomass of the population in stanza *s* is calculated as the proportion of survivors within stanza *s* to the entire age structure (Eq. (13); Christensen and Walters, 2004). EwE calculates *amax*, the maximum age, at 0.9 W_∞ and accounts for the additional biomass in a plus group. We decided that calculation was unnecessary with the current computing power and decided to calculate it at 0.99 W_∞.

$$\frac{a}{\sum_{a=0}^{n} a} = \frac{l_a}{\sum_{a=0}^{n} l_a} \tag{11}$$

$$l_a = e^{-\sum^{a} Z_a - aBAB} \tag{12}$$

$$b_s = \frac{\sum_{a=a_{s,\min}}^{a_{s,\max}} l_a W_a}{\sum_{a=1}^{a_{\max}} l_a W_a} \tag{13}$$

Once the proportional biomass for each stanza is determined, the biomass is calculated using the leading stanza. First the total biomass for the complex is calculated as Eq. (14). Then the trailing stanza groups' biomasses are calculated as Eq. (15). Consumption for each stanza is calculated in a similar way using the consumption by weight assumption of Eq. (16) instead of W_a in Eq. (13). Consumption is then

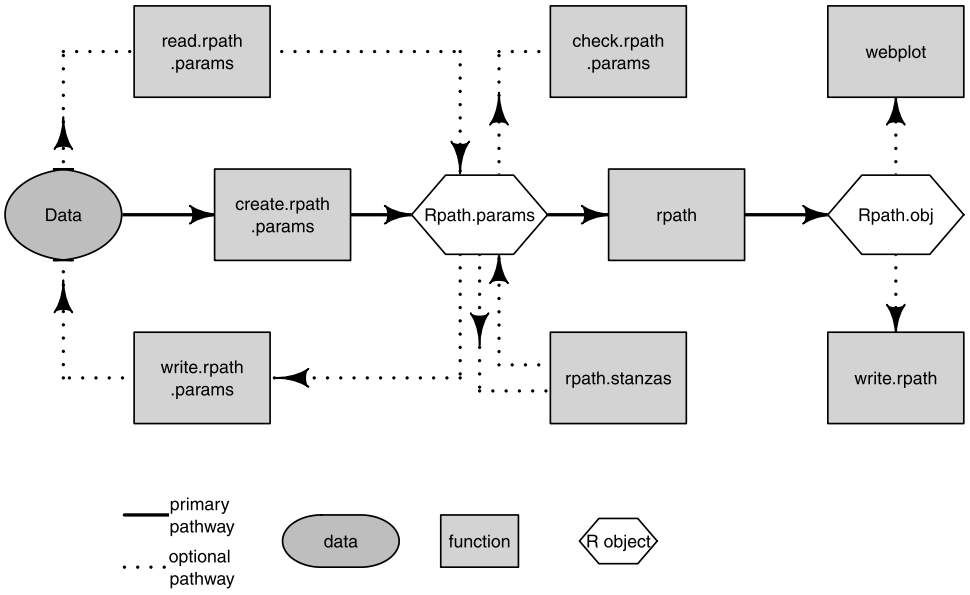


Fig. 1. Pathway of functions for Rpath's Ecopath routines. Data is entered into an *Rpath.params* object via the function *create.rpath.params* or *read.rpath.params*. If multistanza groups are present, the function *rpath.stanzas* will calculate the biomass and consumption for all non-leading stanzas. The *Rpath.params* object can be checked for consistency via *check.rpath.params*. The parameters object can be saved as a series of flat files using *write.rpath.params* or as an R object using the base R function *save*. Once the parameters object is ready, the mass balance is carried out using the function *rpath*. The resultant *Rpath* object can be saved by *write.rpath* or the flow diagram can be visualized using *webplot*.

substituted in for biomass in the calculations of total consumption for the complex Eq. (14) and consumption for trailing stanza groups Eq (15).

$$B_{ms} = \frac{B_{leadings}}{b_{leadings}} \tag{14}$$

$$B_s = b_s B_{ms} \tag{15}$$

$$Q_a = W_a^d \tag{16}$$

Although not used in the mass balance portion of Rpath, *rpath.stanzas* also calculates the abundance at age for each stanza group to be used in dynamic simulations. Abundance at age is based on the relative contribution of recruits by the leading stanza Eq. (17). Actual numbers are calculated as, $N_a = R_i l_a$.

$$R_i = \frac{B_{leading}}{\sum_{a=a_{leading,\min}}^{a_{leading,\max}} l_a W_a} \tag{17}$$

3. Dynamic simulations in Rpath

3.1. From path to rates, Rsim

The static snap shot provided by the Ecopath equations was initially extended to dynamic simulations (Ecosim) by Walters et al. (1997) with later modifications described in Walters et al. (2000) and Christensen and Walters (2004) as incorporated into EwE. Ecosim describes the dynamics of the ecosystem as a set of ordinary differential equations governing the change in the biomass of each group over time and are implemented in Rpath as the Rsim family of functions. Similar to the Rpath and multistanza parameters, the parameters for the Rsim functions can be found in Table 4. For each living (non-detrital) group, the governing equation is:

$$\frac{dB_i}{dt} = (1 - A_i - U_i)\sum_j Q(B_i, B_j) - ForcedMort_{im} \times \left(\sum_j Q(B_j, B_i) - M0_i B_i\right) + ForcedMigrate_{im} \times B_i - C_m(B_i) \tag{18}$$

where $Q(B_i, B_j)$ is the predator/prey functional response between group i and group j, described in Eq. (19), and $C_m(B_i)$ is the time-dependent

Table 3
Pameters used in association with multi-stanza calculations.

Symbol	Units	Parameter	Description
a	number	Age	Age in months of a stanza group
W	kg	Weight	Mass of a group
$k(K_{sp})$	$kgkg^{-n}day^{-1}$	Energy Loss Constant	Derived from the von Bertalaffy growth equation; K_{sp} is the "specialized" form
d	dimensionless	Allometric slope of consumption	Derived from the von Bertalaffy growth equation
t	yr	Time	Time step
l	yr^{-1}	Population Growth Rate-corrected survivorship	Percentage of individuals that survive to the next time step
β	dimensionless	Growth parameter	Slope of the adult growth rate
Se	dimensionless	Spawning energy	Relative amount of energy applied to recruitment rather than growth
Sx	dimensionless	Spawning X	Relative strength of the stock-recruitment relationship
R_{scale}	dimensionless	Recruitment scale	Scaling parameter used to convert adult metrics to juvenile metrics
R_0	Nyr^{-1}	Recruitment rate	Baseline recruitment rate from Rpath
R_p	dimensionless	Recruitment power	Parameter to allow non-linear effects on recruitment
Z	yr^{-1}	Total mortality	Rate of total loss of biomass as a sum of natural mortality and fishing mortality
BA	yr^{-1}	Biomass accumulation	Rate of change of biomass from initial balance
b	proportion	Relative biomass	Percentage of biomass within a given stanza relative to the whole
N	count	Number	Abundance
R	count	Recruits	Number of new group members that become part of the age-1 stanza
vBM	$kgkg^{-n}day^{-1}$	von Bertalaffy growth coefficient	Energy required to grow a unit of weight relative to W_∞ of 1

Table 4
Pameters used in Rsim equations.

Symbol	Rsim code variable	Default value[a]	Description
B_i^*	*B_BaseRef*	*B*	Reference biomass
$M0_i$	*MzeroMort*	*M0*	Rate of "unaccounted" mortality
U_i	*UnassimRespFrac*	*U*	Proportion of consumed food directed to detritus
A_i	*ActiveRespFrac*	*1 - GE - U*	Proportion of consumed food lost to respiration
$Fadj_i$	*FtimeAdj*	0	Range [0–1], foraging adjustment that reflects behavior to change foraging to avoid high predation
QB_i^*	*FtimeQBOpt*	*QB*	Optimal QB for foraging time adjustments
H_i	*HandleSelf*	0	Range [0–1], determines whether handling time is calculated across range of prey (1) or for each individual prey (0)
S_i	*ScrambleSelf*	0	Range [0–1], determines whether predator density dependence is calculated across range of predators (1) or for each individual predator (0)
Q_{ij}^*	*QQ*	*QB*B*DC*	Reference consumption rate between each predator/prey pair
D_{ij}	*DD*	1000	Range [1 - ∞], handling time curvature of functional response
V_{ij}	*VV*	2	Range [1 - ∞], vulnerability curvature of functional response
α_{kj}	*PredPredWeight*	$\frac{2Q_{ik}^* V_{ik}}{(V_{ik}-1)B_i^* B_k^*}$	Relative weight of predators in pooled density dependence response
β_{ik}	*PreyPreyWeight*	*DC*	Relative weight of prey in pooled handling time response
θ_{ij}	*HandleSwitch*	*M0*	Range [0 - 2], controls shape of functional response (prey density dependence)
q_{ig}	*FishQ*	$\frac{C_{ig,land}^*}{B_i^*}, \frac{C_{ig,disc}^*}{B_i^*}$	Catchability of each species by each gear type
detfate	*DetFrac*	*detfate*	Proportion of detritus from each group directed to each detritus group
	COUPLED	1	If 0, HandleSwitch is assumed to be 0 for all predator/prey pairs (density-independent feeding)
	RK4_STEPS	4	Number of steps in 1 month for RK4 integration method

[a] Value from corresponding variable in Ecopath model unless otherwise specified

fisheries-induced mortality function described in Eq. (25). A_i and U_i are energetic loss terms, specified as a fraction of total consumption, representing active respiration loss and unassimilated food, respectively. By default, $(1 - A_i - U_i)$ is equal to the Ecopath growth efficiency (*GE*; Eq. 2). The active respiration portion of consumption is considered "lost" to the system, while the unassimilated portion is gained by detrital groups in proportion to each living group's Ecopath detrital fate Eq. (3). $M0_i$ is a mortality term for "other" mortality, with a default value calculated from Ecopath Eq. (9); the total biomass lost from a living group due to "other" mortality also enters detritus according to the group's detrital fate. $ForcedMort_{im}$ and $ForcedMigrate_{im}$ are forcing functions described further in Section 3.4.

3.2. Predator-prey responses

For the predator/prey functional response $Q(B_i, B_j)$, Rpath uses an alternate form of the Walters et al. (1997) consumption equation derived in Aydin (2004), in order to better relate the functional form to fitting and uncertainty analysis (e.g. Gaichas et al., 2011; Gaichas et al., 2012). The formulation separates density-dependent effects for predator j and prey i using:

$$Q(B_i, B_j) = Q_{ij}^* \left(\frac{V_{ij} Ypred_j}{V_{ij} - 1 + (1 - S_{ij}) Ypred_j + S_i \sum_k (\alpha_{kj} Ypred_k)} \right) \times \left(\frac{D_{ij} Yprey_i^{\theta_{ij}}}{D_{ij} - 1 + ((1 - H_{ij}) Yprey_i + H_i \sum_k (\beta_{ik} Yprey_k))^{\theta_{ij}}} \right) \quad (19)$$

The state variables in Eq. (19) are contained in the terms *Ypred* and *Yprey*, that represent each species' foraging activity as predator and prey respectively. These are calculated as ratios with respect to set reference levels:

$$Ypred_j = ForcedPred_{jm} \times Ftime_j \times B_j / B_j^* \quad (20)$$

$$Yprey_i = ForcedPred_{im} \times Ftime_i \times B_i / B_i^* \quad (21)$$

$Ftime_i$ is a state variable that has an initial condition of 1.0 and does not change unless the foraging time adjustment parameter, $Fadj_i$ is nonzero. Therefore, at Ecopath equilibrium, *Ypred* and *Yprey* are equal to 1, so that Eq. (19) reduces to $Q(B_i, B_j) = Q_{ij}$. *ForcedPred* is a forcing function explained in more detail in Section 3.4. Primary producers are modeled as predators using Eq. (19), but with $Yprey_i$ (i.e. "sunlight") fixed at 1.0. Therefore, the self-saturation of phytoplankton group j is set through V_{ij}.

$$Ftime_{i,m+1} = 0.1 + 0.9 \times Ftime_{im} \left(1 + Fadj_i \left(\frac{QB_i^*}{QB_{im}} - 1 \right) \right) \quad (22)$$

Changes in *Ftime* are made on a monthly time step m using the difference Eq. (22) where QB_i^* is the reference consumption to biomass ratio (from Ecopath by default) and QB_{im} is the predator's current total consumption summed across all prey Eq. (19), divided by its current biomass (Walters and Kitchell, 2001). Importantly for comparison, in Rpath, the default for $Fadj_i$ is 0, while in EwE the default is 0.5, leading to differences in results when using the default assumptions from respective implementations. The higher value will result in a smaller magnitude perturbation as species groups consume more to mitigate their losses.

In Eq. (19), Q_{ij}^* is the reference consumption rate (consumption flow from prey i to predator j). V_{ij} is the main "vulnerability" parameter input into Ecosim to account for predator density dependent foraging (X_{ij} in the original derivation of Walters et al. (1997)) and D_{ij} is the "handling time" parameter accounting for prey density dependence (predator saturation). The range of V_{ij} and D_{ij} is [1, ∞]; for practical use, values > 1000 are indistinguishable from infinity. The default value for V_{ij} is 2.0, while the default value for D_{ij} is 1000 (Table 6). θ_{ij} controls the shape of the prey density dependent response, with a value of 1.0 (default) resulting in a Holling Type II functional response, and a value of 2.0 representing a Holling Type III response, though θ_{ij} is not restricted to integer values.

The parameters S_{ij} and H_{ij} are flags that control whether the predator density dependence (S_{ij}) or prey density dependence (H_{ij}) are affected solely by the biomass levels of the particular predator and prey, or whether a suite of other species' biomasses in similar roles impact the relationship. For the default value for S_{ij} of 0 (off), the predator density dependence is only a function of that predator biomass and likewise for prey with the default value of 0 for H_{ij}. Values greater than 0 allow for a density-dependent effects to be affected by a weighted sum across all species for predators Eq. (23), and for prey Eq. (24). The weights α_{kj} and β_{kj} are normalized such that the sum for each functional response (i.e. $\Sigma_k \alpha_{kj}$ and $\Sigma_k \beta_{kj}$ for the functional response between predator j and prey i) sum to 1. The weights are calculated from the density-independent search rates for each predator/prey pair, (a_{ij} terms in Walters et al., 1997), which is equal to $2Q_{ij}^* V_{ij} / (V_{ij} - 1) B_i^* B_j^*$.

Table 5
List of available forcing parameters implemented in Rsim. All forcing parameters can be adjusted using the *adjust.forcing* function except for *ForcedEffort, ForcedFrate,* and *ForcedCatch* which are modified using the *adjust.fishing* function.

Parameter	Description	Default
ForcedEffort	Modifies the relative fishing effort (E_{gm}) associated with a particular fleet	1
ForcedFrate	Multiplier to biomass that is then added to loss due to fishing (Not associated with a particular fleet)	0
ForcedCatch	Added directly to loss due to fishing (Not associated with a particular fleet)	0
ForcedPrey	Modifies effective prey biomass resulting in increased predation	1
ForcedSearch	Modifies effective predator biomass resulting in increased consumption	1
ForcedMort	Modifies loss due to predation and "other" factors resulting in a decrease in biomass not accounted for by any predator or fleet	1
ForcedRec	Modifies egg production	1
ForcedMigrate	Multiplier to biomass that is then added as a loss term (Positive value indicate emmigration while negative values indicate immigration)	0
ForcedBio	Overrides biomass dynamics from simulation	-1

$$PredSuite = \sum_k (\alpha_{kj} Ypred_k) \quad (23)$$

$$PreySuite = \sum_k (\beta_{ik} Yprey_k) \quad (24)$$

3.3. Fishing responses

A primary use of Ecosim is to simulate both the direct and indirect effects of fishing on ecosystem biomasses. Eq. (25) offers multiple ways to specify time-specific catch:

$$C_m(B_i) = \sum_{g=1}^{n} q_{ig,land} E_{gm} B_i + \sum_{g=1}^{n} q_{ig,disc} E_{gm} B_i + ForcedFRate_{im} B_i + \min(ForcedCatch_{iy}/12,\ B_i + P_{i,net}) \quad (25)$$

Here, $q_{ig,land}$ and $q_{ig,disc}$ are calculated for each fleet g by assuming that E_{gm} is 1.0 in Ecopath equilibrium, so that q is calculated from Ecopath catches as Eqs. (26) and 27. The gear effort E_{gm} can be supplied by the user at a monthly resolution m, and defaults to 1.0. $ForcedFrate_{im}$ is an additional instantaneous fishing mortality rate (defaulting to 0) that can be applied to a functional group independently of gear type. While forcing rates are input at a monthly resolution, the units are annual (year^{-1}). An additional, absolute loss term, $ForcedCatch_{iy}$ can be further supplied at an annual resolution (as total catch, in biomass units, for the simulation year, defaulting to 0). This annual quantity is divided into 12 equal monthly catches. As the loss term *ForcedCatch* is not dependent on biomass, if the loss would exceed the biomass of a group plus its production in a particular time step ($B_i + P_{i,net}$), then the removal is set to ($B_i + P_{i,net}$), this may cause the biomass of the group to crash and prevent the desired catch from being achieved (although the desired $ForcedCatch_{iy}$ is input with annual resolution, catches for later months in the year are not increased to make up for catch shortfalls in earlier months). Catch removed through gears retains the discard rate of those gears; that is, the proportion of catch discarded by each gear input into Ecopath enters detrital groups according to each gear's detrital fate Eq. (3). All catch removed through *ForcedFrate* or *ForcedCatch* are considered "landings" and removed from the system and do not enter a detrital group.

$$q_{ig,land} = \frac{C^*_{ig,land}}{B_i^*} \quad (26)$$

$$q_{ig,disc} = \frac{C^*_{ig,disc}}{B_i^*} \quad (27)$$

Table 6
List of default values for rsim parameters.

Parameter	Default
E	1
d	2/3
V	2
S	0
D	1000
H	0
θ	1
Se	1
Sx	1000
Rp	1
R_{scale}	1

3.4. Other forcing responses

In addition to fishing, Rpath offers several methods of applying time-dependent data ("forcing") to an Rsim run, to account for various sources of environmental variability (Table 5). In Eq. (18), *ForcedMort* and *ForcedMigrate* are forcing functions for mortality and migration, respectively. In Eq. (20), *ForcedPred* affects the predation rate of a predator across all its prey, while in Eq. (21) *ForcedPrey* affects the exposure of prey to all of its predators. *ForcedRec* affects the egg production in Eq. (36). All environmental forcing functions are input with monthly resolution m.

ForcedMort, ForcedPred, ForcedPrey, and *ForcedRec* are non-negative multipliers, with a default value of 1.0 for all months of the simulation. Mortality forcing affects both losses by predation and "other" losses by multiplying *ForcedMort* to the sum of predation consumption as well as M0 Eq. (18). *ForcedMigrate* is input with monthly resolution as an annualized instantaneous rate (units of year^{-1}); it is applied as a loss term so positive values indicate emigration from the system while negative values indicate immigration into the system.

Finally, Rpath allows the biomass dynamics equations of Rsim to be overridden on a species-by-species level by entering monthly time series of biomass into the *ForcedBiomass* forcing function; if *ForcedBiomass* contains a non-negative value for species i and time step m, the biomass of that group is set equal to the input value for the next time step.

3.5. Setting the scene, rsim.scenario

While the Ecosim equations can be parameterized independently of Ecopath, the supported use in both EwE and Rpath is to use a balanced Ecopath model to create an initial Ecosim parameter set. In Rpath, the initial parameterization is accomplished using the *rsim.scenario* function (Fig. 2). The initial biomass for each group (biomass at t = 0, hereafter B_i^*) is taken from the balanced biomass of Eq. (1). *GE* and *M0* are taken from Eqs. 2 and 9 respectively, and fisheries catch is assumed to have a constant Ecopath-calculated mortality rate Eq. (8). The parameters for each predator/prey functional response $Q(B_i, B_j)$ are set such that if $B_i = B_i^*$ and $B_j = B_j^*$, then $Q(B_i, B_j) = M2_{ij} B_i = DC_{ij} QB_j B_j B_i$ Eq. (7). With this parameterization, if the Ecopath model is balanced (all *EEs* < 1.0) and all biomass accumulation and/or emigration is 0, the system of dynamic equations implied by Eq. (18) has a stable equilibrium ($\frac{dB_i}{dt} = 0$ for all i) at its initial state where all $B_i = B_i^*$. If biomass accumulation

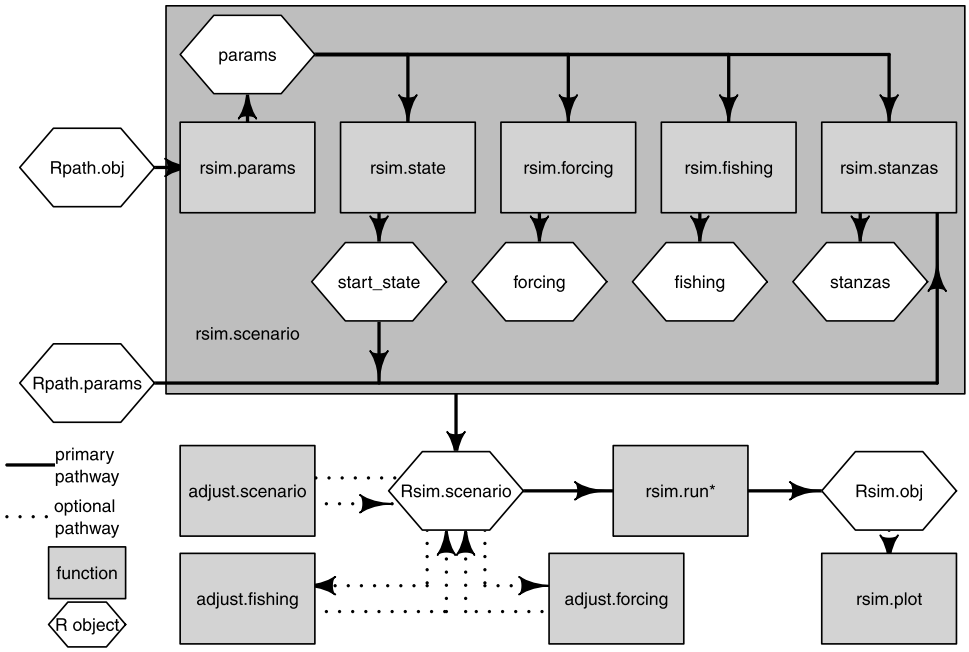


Fig. 2. Pathway of functions for Rpath's Ecosim routines. The Ecosim routines require *Rpath* and *Rpath.params* objects. The *rsim.scenario* function generates all of the rates and internal list objects necessary to run *Rsim* and bundles them within an *Rsim.scenario* object. Within *rsim.scenario* there are five sub-functions: *rsim.params, rsim.state, rsim.forcing, rsim.fishing,* and *rsim.stanzas*. The resultant list objects can be adjusted using the three adjust functions before being passed to the *rsim.run* function. The *rsim.run* function contains C++ code. The simulation run can be visualized using the *rsim.plot* function.

and emigration have a net nonzero value, or a group has an $EE > 1.0$, then the derivative $\frac{dB_i}{dt}$ will be out of equilibrium at $B_i = B_i^*$ by that amount.

Sub-functions of *rsim.scenario* generate default list objects that run the simulations (Fig. 2). The sub-function *rsim.state* creates numeric vectors for reference biomasses from the balanced ecopath model, numbers for multi-stanza groups, and foraging time. The sub-function *rsim.fishing* creates three matrices. The first matrix has a number of columns equal to the number of fleets and contains time varying fishing effort by month. The other two matrices have a number of columns equal to the number of species groups and a number of rows equal to the length of the simulation in years. These two are used for *ForcedFrate* and *ForcedCatch*. The sub-function *rsim.forcing* generates six matrices corresponding to the remaining forcing functions (Table 5). Each has a number of columns equal to the number of species groups and a number of rows equal to the length of the simulation in months.

The sub-function *rsim.stanzas* generates the parameter list necessary to perform multi-stanza calculations. Unlike the other sub-functions, a large part of this routine is carrying over the appropriate numbers from the Rpath.param object rather than default values (Fig. 2). The sub-function accounts for the species group codes, beginning and end months for each stanza, as well as the weight, number, and consumption at age calculated using the *rpath.stanzas* function. It also sets up the base spawning biomass Eq. (28) and the slope of the adult growth rate Eq. (29). For Eq. (29), *vBM* is the energy required to grow a unit in weight scaled to W_∞ from the von Bertalanffy equation, $vBM = 1 - 3(K_{sp}/12)$ where *Ksp* is divided by 12 to convert from yearly to monthly value.

$$B_{spawn} = \sum_{W_{as} > W_{mat}} (N_{as}(W_{as} - W_{mat})) \tag{28}$$

$$\beta_a = W_{a+1} - vBM \times W_a \frac{\sum (N_{as} Q_{as})}{Q_s} \tag{29}$$

3.6. Adjustment functions

While *rsim.scenario* creates a list object that is populated by default values (Table 6), all of the parameters can be modified for a given scenario. Users can choose to modify the parameter lists directly but that is not recommended. For efficiency, the numerical integration is done in C++ rather than R. However, R and C++ index differently which can cause some confusion. To compensate, all of the vectors and matrices created for the simulations have a leading zero. This can make it difficult to determine which value should be modified for a given scenario. Therefore in order to assist users, a series of adjustment functions have been developed (Fig. 2). General parameters generated by *rsim.params* can be adjusted using the function *adjust.scenario*. The function *adjust.fishing* can change fishing effort (*ForcedEffort*), *ForcedFrate, or ForcedCatch*. Finally, *adjust.forcing* will modify the matrices created by the *rsim.forcing* sub-function. These functions will ensure that the correct inputs are changed prior to running the simulation. More on how to use the adjustment functions can be found in the Rsim vignette that accompanies the Rpath package.

3.7. Running rsim

The differential equations of the Ecosim routine can be numerically integrated using either a 4th order Runge-Kutta (RK4) or a two-step Adams-Bashforth (AB) method. The time step for the AB method is fixed at one time step per month. The RK4 method defaults to four time steps per month, but this can be altered by the user. The trade-off between the two methods is accuracy and speed – RK4 requires four derivative calculations per time step (Fig 3) or, with the default time step, 16 calculations per month, while AB requires one plus the derivative saved from the previous time step (Fig 4). With a single monthly time step, the AB method is too coarse to model rapid seasonal-scale changes (e.g. plankton blooms). To provide stability for modeling inter-annual changes in species with high turnover rates, the AB method includes the "fast equilibrium" method described in Walters et al. (1997); by default, this is used in the AB method for species with a turnover rate greater than once a month as determined by $M0B_i > 24$. Changes in biomass for those groups are calculated as Eq. (30). Where *TotalGains* are a summation of biomass gained via consumption and *TotalLoss* the sum of biomass loss to predation, fishing, and "other" sources. Press et al. (2007) offers more details on the two methods.

$$B_{i,t} = 0.5 \frac{TotalGains}{TotalLoss/B_i^*} + 0.5 B_{i,t-1} \tag{30}$$

Both numerical integration methods use the same underlying sub-functions. There is a sub-function that sums up multi-stanza groups to total biomass, one that calculates the derivative of the biomass at each

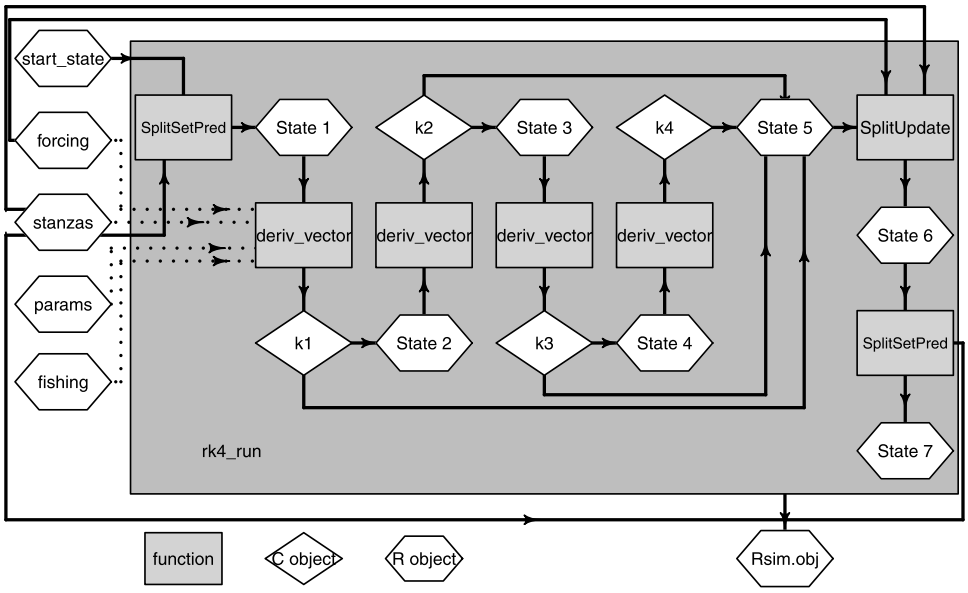


Fig. 3. Pathway of functions for the *rsim.run* sub-function *rk4_run*. This is the 4th order Runge-Kutta routine as described in Press et al. (2007). The function is intialized using the *Rsim.scenario* object (not shown) which includes the *start_state, fishing, params, forcing*, and *stanzas* objects. All steps inside the *rk4_run* function are carried out using the Rcpp package. Models without multi-stanza groups bypass the *SplitSetPred* and *SplitUpdate* functions. Subsequent time steps iterate using the final state as the initial state (State 1).

time step, and one that updates the multi-stanza parameters. The first sub-function, *SplitSetPred*, sums the biomass, consumption, and numbers for each multi-stanza group at time step t as Eqs. (31), (32), and (33).

$$B_t = \sum_{a=1}^{n} (N_a W_a) \tag{31}$$

$$Q_t = \sum_{a=1}^{n} (N_a Q_a) \tag{32}$$

$$N_t = \sum_{a=1}^{n} N_a \tag{33}$$

The sub-function *deriv_vector* calculates the change in biomass, consumption, and fishing at each time step. Changes in biomass are based on the total gains by a species group offset by the total losses. Total gains are a sum of the food consumed Eq. (19) while losses are a combination of predation Eq. (19), fishing Eq. (25), energetics (respiration and unassimilated food), and "other" losses Eq. (18).

The final sub-function of *rsim.run* is *SplitUpdate*. This sub-function updates all of the multi-stanza parameters before the next time step. The survival rate is updated base on the actual loss of biomass and applied to the abundance at age Eq. (34). Weight at age is updated as Eq. (35). Spawning biomass is then updated as Eq. (28). Abundance of the oldest age group is a sum of the surviving members of the age group and those surviving to the plus group from the previous age group. Weight-at-age for the plus group is an average of the previous plus group's weight-at-age and the updated weight-at-age for the younger age group. All other age groups are updated from the younger age group's survivors and weights. Finally, the relationship between recruits and spawners is determined using Eq. (36) and scaled in Eq. (37).

$$N_a = N_a e^{-\left(\frac{B_{s,loss}}{12}/B_s\right)} \tag{34}$$

$$W_a = W_a(vBM_i) + GE(\beta_a) \tag{35}$$

$$Egg = \frac{B_{sp}(Se)Sx}{Sx - 1 + \frac{B_{sp}}{B_{sp}^*}} \times ForcedRec \tag{36}$$

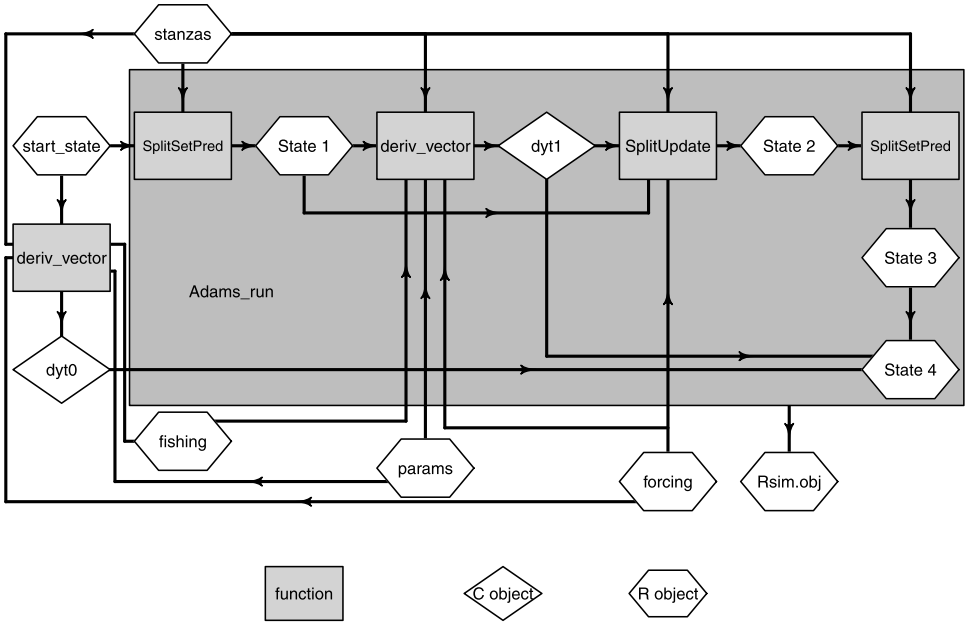


Fig. 4. Pathway of functions for the *rsim.run* sub-function *Adams_run*. This is the Adams-Bashforth routine as described in Press et al. (2007). The function is intialized using the *Rsim.scenario* object (not shown) which includes the *start_state, fishing, params, forcing*, and *stanzas* objects. There is also an initial derivative calculated before running the C++ code. All steps inside the *Adams_run* function are carried out using the Rcpp package. Models without multi-stanza groups bypass the *SplitSetPred* and *SplitUpdate* functions. Subsequent time steps iterate using the final state as the initial state (State 1) and *dyt1* as *dyt0*.

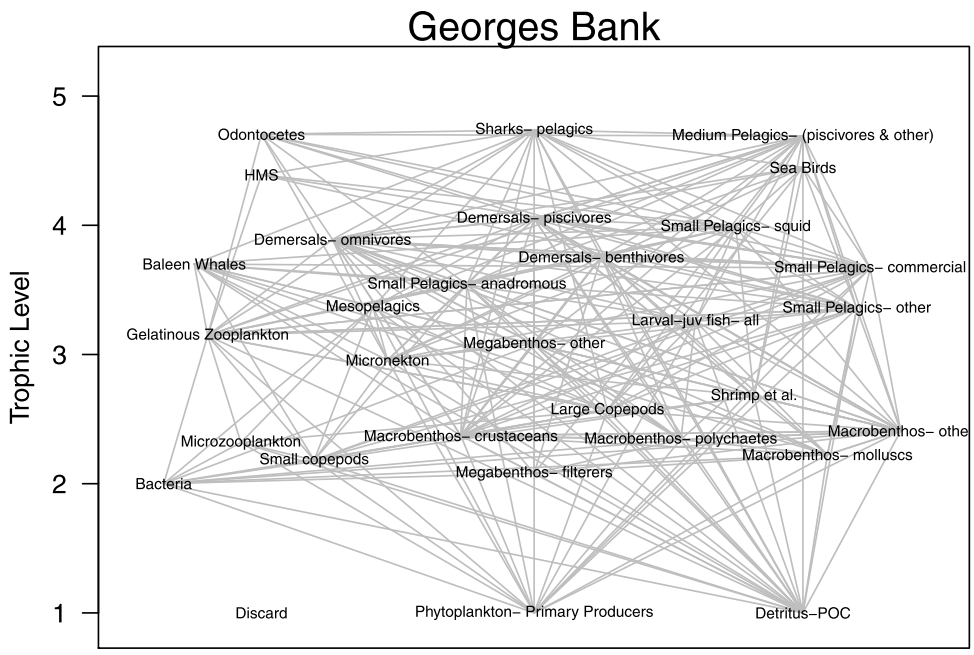


Fig. 5. An example of the output from the webplot function. This is the Georges Bank model from Link et al. (2006).

$$N_{a=1} = R_{scale} R_0 (\frac{E}{E^*})^{Rp} \quad (37)$$

4. Graphical routines

One advantage of implementing EwE on the R platform is the ability to take advantage of R's built-in graphical engine. There are many graphical packages available for R, and even the base package can produce publication quality figures. We included two built-in graphical routines that can be applied to the outputs from the functions rpath and rsim.run. The first is a function called webplot which uses an *Rpath.obj* generated by rpath. This creates a foodweb similar to the flow diagrams available in the EwE software (Fig. 5). The second is a function for visualizing the results of rsim simulations called *rsim.plot* (Fig. 6). For more details on how to customize the outputs see the vignettes that are included with the Rpath package.

5. Discussion

As ecosystem-based fisheries management (EBFM) moves from the theoretical stage towards implementation, the need for flexible modeling tools will increase. Ecopath with Ecosim is a popular tool for ecosystem modelers and offers an open source platform (Coll et al., 2015; Heymans et al., 2016; Steenbeek et al., 2015). Rpath is the latest extension of the EwE Windows software to another code base. Fortran and Matlab versions are also available (Akoglu et al., 2015; Kearney), and both provide flexibility to extend beyond the traditional scope of the EwE software. Like those versions, our R implementation is not meant to be a replacement to the current EwE software but was

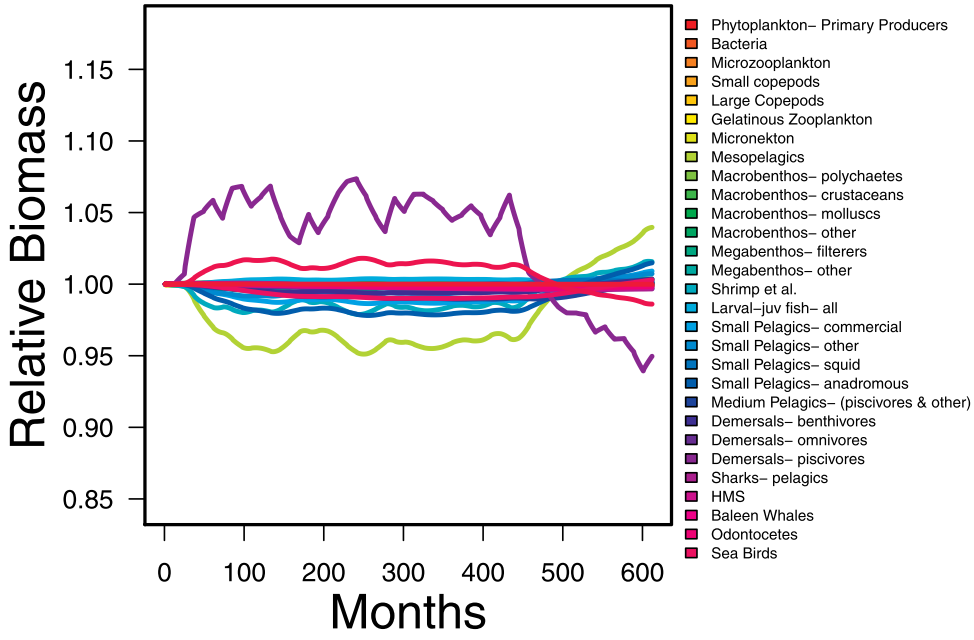


Fig. 6. Example output of the *rsim.plot* function. This is a simulation run using the Georges Bank ecosystem from Fig. 5 (unpublished).

developed as a complement. Rpath provides two services above the standard EwE software: reproducibility and adaptability.

Although replication is the gold standard in science, it is nearly impossible for ecosystem modeling. This common problem facing many computational science fields has led to a call for reproducible research (Peng, 2011). EwE's greatest strength is its easy to use GUI, but using a complex GUI may lead to user error, hampering the ability to reproduce the research. Users can use the base code directly in a script if so desired to avoid this potential pit fall (Steenbeek et al., 2015). Likewise, R is adept at providing reproducibility, because the code needed to run analyses can be stored in a script and made publicly available via services like GitHub, allowing reviewers and other researchers the chance to recreate a given model or simulation.

The streamlined work flow of Rpath using a single software platform enhances reproducibility. At the 30th anniversary EwE conference in Barcelona, Spain (Coll et al., 2015), several presenters referenced R or one of its packages for some part of their analysis. Several presenters used an R package called EcoTroph, which was developed as an EwE add-on to conduct trophic network analysis (Colléter et al., 2013). Rpath offers users the ability to keep all their analysis in one place rather than switch between software platforms.

The open source nature of EwE allows for the modification of the code to fit the user's needs. Rpath expands on the open source nature of the code by porting it to a commonly used computer dialect by ecologists, R. Some of the underlying simulation code is written in C++ for speed. The C++ code is available for modification, but it may require a more advanced programmer to modify. Fortunately, the individual functions within the C++ code are available within R which allows them to be used independently if so desired.

The current version of Rpath only contains the base functionality of EwE. This excludes the spatial module, Ecospace, as well as several more minor features that have been developed over time for specific use cases. However, with the amount of interest generated by Rpath, we anticipate community development and expect that more advanced functions of EwE will be coded within the Rpath framework. Considering the flexibility of R, new developments can include functionality that is not part of the EwE software. For example, there is active development on a Bayesian estimation approach based on parameter re-sampling called Ecosense (See, e.g. Gaichas et al., 2015).

Our goal for this article was to make the Rpath code as transparent as possible. The future of Rpath relies on community development which requires an understanding of the underlying mechanics. Rpath is not meant for the novice EwE user but to allow more advanced users the opportunity to tailor their models and outputs to their needs. R offers a wide range of statistical and graphical packages which can be used to enhance mass balance studies. As a community, the impetus will be to share advancements to the modeling framework and unlock the full potential of the open source nature of EwE. Further development of this flexible tool that integrates statistical analysis and visualization tools in one package will be extremely useful in bridging the gap from ecosystem management theory to practice.

CRediT authorship contribution statement

Sean M. Lucey: Methodology, Writing - original draft, Software, Validation. **Sarah K. Gaichas:** Methodology, Writing - review & editing, Software. **Kerim Y. Aydin:** Methodology, Writing - review & editing, Software, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.ecolmodel.2020.109057.

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